As per Quest 8-1

27-08-2020

100.0mm

100.0mm

B-Cobal Film-Coated Tablet

Mecobalamin 0.5mg, Vitamin B1 100mg, and Vitamin B6 200mg.

Product Description

A round, red film-coated tablet with diameter about 12mm.

Pharmacodynamics

Actions:

B-cobal is a combination of the 3 essential neurotropic vitamin B group.

Thiamine (Vit B1) is involved in carbohydrate metabolisme and in nerve transmission. Pyridoxine (Vit B6) is involved in amino acid metabolism and in protein metabolism. Thiamine combines with adenosine triphosphate (ATP) to form a coenzyme, thiamine pyrophosphate (thiamine diphosphate, cocarboxylase), which is necessary for carbohydrate metabolism.

The body needs pyridoxine hydrochloride in order to make cells of the immune system. Pyridoxine hydrochloride helps to improve immune response to the increase in production of antibodies and also helps in communicative interactions between cytokines and chemokines.

Mecobalamin is a B12 containing co-enzyme with an active methyl base. It participates in transmethylation reactions and is the most active of all B12 homologs in the body with respect to nucleic acid, protein and lipid metabolism.

Mecobalamin acts to repair damaged nerve tissue in nerve disorders eg: axonal degeneration and demylination; and it is involved in erythroblast maturation, promotion of erythroblast division, and heme synthesis, thus acting to improve the status of the blood in megablastic anemia.

Vitamin B1, B6, B12 are indispensable to normal nerve cell metabolism and this combination greatly enhances their therapeutic efficacy as compared to the effect of the respective factors given individually.

Pharmacology

(1) *Mecobalamin is a kind of endogenous coenzyme B12:*
As a coenzyme of methionine synthetase, mecobalamin plays an important role in transmethylation in the synthesis of Methionine from homocysteine.

(2) *Mecobalamin is well transported to nerve cell organelles, and promotes nucleic acid and protein synthesis:*
Experiments in rats show that mecobalamin is better transported to nerve cell organelles than cyanocobalamin and promotes nucleic acid and protein synthesis more than cobamamide does. Experiments with cells from the brain origin and spinal nerve cells in rats also show mecobalamin to be involved in the synthesis of thymidine from deoxyuridine, promotion of deposited folic acid utilization and metabolism of nucleic acid.

(3) *Mecobalamin promotes axonal transport and axonal regeneration:*
In rat models with streptozotocin-induced diabetes mellitus, mecobalamin normalizes axonal skeletal protein transport in sciatic nerve cells. Mecobalamin exhibits neuropathologically and electrophysiologically inhibitory effects on nerve degeneration in neuropathies induced by drugs, such as adriamycin, acrylamide, and vincristine (in rats and rabbits), models of axonal degeneration in mice and neuropathies in rats with spontaneous diabetes mellitus.

(4) *Mecobalamin promotes myelination (phospholipid synthesis):*
Mecobalamin promotes the synthesis of lecithin which is the main constituent of medullary sheath lipid. It also increases myelination of neurons in rat tissue culture more than cobamamide does.

(5) *Mecobalamin restores delayed synaptic transmission and diminished neurotransmitters back to normal:*
Mecobalamin restores end-plate potential induction early by increasing nerve fiber excitability in the crushed sciatic nerve in rats. In addition, mecobalamin normalizes diminished levels of acetylcholine in brain tissue of rats fed with a choline-deficient diet.

Pharmacokinetics

(1) *Absorption*
Vitamin B12 substances bind to intrinsic factor, a glycoprotein secreted by the gastric mucosa, and are then actively absorbed from the gastrointestinal tract. Absorption is impaired in patients with an absence of intrinsic factor, with a malabsorption syndrome or with disease or abnormality of the gut, or after gastrectomy. Absorption from the gastrointestinal tract can also occur by passive diffusion; little of the vitamin present in food is absorbed in this manner although the process becomes increasingly important with larger amounts such as those used therapeutically.

(i) *Single-dose administration*
When mecobalamin was administered orally to healthy adult male volunteers at single doses of 120 µg and 1,500 µg during fasting, the peak serum total vitamin B12 concentration was reached after 3 hrs for both doses, and this was dose-dependent. (Note) A single dose of 1,500 µg is unapproved.

(ii) *Repeated-dose administration*
When mecobalamin was administered orally to healthy adult male volunteers at a dose of 1,500µg daily for 12 consecutive weeks, the serum concentration increased for the first 4 weeks after administration, rising to about twice as high as the initial value. Thereafter, there was a gradual increase which peaked at about 2.8 times the initial value at the 12th week of dosing. The serum concentration declined after the last administration (12 weeks), but was still about 1.8 times the initial value 4 weeks after the last administration.

(2) *Distribution*
Vitamin B12 is extensively bound to specific plasma proteins called transcobalamins; transcobalamin II appears to be involved in the rapid transport of the cobalamins to tissues. Vitamin B12 is stored in the liver. Vitamin B12 diffuses across the placenta and also appears in breast milk.

(3) *Excretion*
Vitamin B12 is excreted in the bile, and undergoes extensive enterohepatic recycling; part of a dose is excreted in the urine, most of it in the first 8 hours; urinary excretion, however, accounts for only a small fraction in the reduction of total body stores acquired by dietary means. 40-80% of the cumulative amount of total B12 excreted in the urine by 24 hrs after single-dose administration was excreted within the first 8 hrs.

(4) *Elimination Half-life (mecobalamin)*
12.5 hrs (single-dose oral administration; calculated from the average of 24-48 hour values)

120.0mm

Thiamine (Vit B1) is absorbed from the gastrointestinal tract and is widely distributed to most body tissues. It is not stored in the body and amounts in excess of the body is requirements are excreted in urine as unchanged thiamine or its metabolite, pyrimidine.

Pyridoxine (Vit B6) is absorbed from the gastrointestinal tract and is converted to the active form: Pyridoxal phosphate. Pyridoxal phosphate is transformed to pyridoxic acid and excreted in the urine.

When mecobalamin administered by oral, peak plasma concentration is achieved after 3 hrs and it is dose-related. The excretion mainly through urine, approximately 40 -80 % of the total urine excretion is excreted within first 8 hours.

Indication

For neurological and other disorders associated with disturbance of the nerve cell metabolism in which high dose B-complex vitamins play a role. These include: Polyneuritis (of toxic and non-toxic etiology): neuralgias eg: lumbago, sciatica, root irritation due to degenerative changes of the vertebral column, shoulder-arm syndrome, herpes zoster, cervicalsyndrome, ischialgia, etc: diabetic neuropathy; metabolic and neuropathic changes due to pregnancy and oral contraceptives; convalescence.

Megaloblstic anemia due to B12 deficiency.

Mode of Administration

Oral

Recommended Dosage

Usually for adults, orally administer 1 tab 3 times a day.

The dosage should be adjusted according to age of patient and severity of symptoms.

Contraindication

Patients with a history of hypersensitive to vitamin B and mecobalamin.

Warning and precautions

This product should not be used aimlessly for more than one month unless it is effective.

Precautions concerning use

Administration: Mecobalamin is susceptible to photolysis. Light decreases the content of mecobalamin and tablets may change colour (eg. turn reddish) with exposure to moisture. Therefore, this product should be used promptly after the package is opened, and caution should be taken so as not to expose the tablets/capsules to light/moisture.

Other Precautions: The prolonged use of larger doses of mecobalamin is not recommended for patients whose occupation requires the handling of mercury or mercury compounds.

Long term administration of large doses of pyridoxine is associated with the development of severe peripheral neuropathies. This may occur with doses in excess of about 2g daily.

Interactions with Other Medicaments

Metformin, histamin H-2 receptor antagonists (cimetidine, ranitidine), omeprazole, aminoglycosides, colchicines, aminosalicylic acid, anticonvulsants, neomycin and alcohol may decrease the absorption of vitamin B12.

Antibiotics, Tetracycline - Vitamin B1 and Vitamin B12 should not be taken at the same time as the antibiotic tetracycline because it interferes with the absorption and effectiveness of this medication. Mecobalamin tablet should be taken at different times from tetracycline.

Antidepressant Medications, Tricyclic - Taking vitamin B1 supplements may improve treatment with antidepressants such as nortriptyline, especially in elderly patients. Other medications in this class of antidepressants include desimpramine and imipramine.

Digoxin - Laboratory studies suggest that digoxin may reduce the ability of heart cells to absorb and use vitamin B1. Diuretics particularly furosemide, may reduce the levels of vitamin B1 in the body.

Pyridoxinie (Vit B6) decreases the effects of the levodopa.

Serum concentrations may be decreased by use of oral contraceptives.

Many of these interactions are unlikely to be of clinical significance but should be taken into account when performing assays for blood concentrations.

Pregnancy and Lactation

Pregnancy

No risks have become known associated with the use of B-cobal during pregnancy at the recommended dosage.

Lactation

Vitamins B1, B6 and B12 are secreted into human breast milk, but risks of overdosage for the infant are not known. In individual cases, high doses of vitamin B6 ie, >600 mg daily, may inhibit the production of breast milk.

Side Effect

Hypersensitivity reactions to vitamin B1 eg, sweating, tachycardia (rapid heartbeat), and skin reactions with itching and urticaria are very rare.

Chromaturia ("reddish urine", appears during the first 8 hours after an administration and typically resolves within 8 hours after an administration and typically resolves within 48 hours) may occur.

Gastrointestinal complaints eg, nausea, vomiting, diarrhoea or abdominal pain may occur.

Symptoms and Treatment of Overdose

Sensorial neuropathy and other sensorial neuropathy syndromes which can be caused by long-term administration of high doses of pyridoxine improve gradually upon vitamin discontinuation.

Storage Conditions

Store below 30°C in an airtight container. Protect from light and moisture.

Shelf-life

Three years from the date of manufacture.

Dosage Forms and Packaging Available

Amber glass bottle contains 30 tablets.

Manufacturer & Product Registration Holder

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Lot 10621 (PT 16700), Jalan Permata 2, Arab Malaysian Industrial Park,
Nilai, 71800 Negeri Sembilan, Malaysia.

Date of Revision of Package Insert

Jul 2020 Ver.09

Keep out of the sight and reach of children.

120.0mm

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